DATA SCIENCE PROJECT

Clustering Solar Energy Production Zones

Identifying High-Potential Solar Regions in New York State through Machine Learning Clustering Analysis

218,115	62	3	0.5065
Solar Projects	Counties Clustered	Production Zones	Silhouette Score

Keerthana | April 2026

PROJECT OVERVIEW

Problem Statement

Leverage data on New York State solar projects to cluster geographical areas by energy production — enabling strategic targeting and data-driven business decisions.

Dataset

218,115 solar projects • 17 columns • NY State • 2000–2023 • 62 unique counties

DELIVERABLES

01 Clustering Algorithm

KMeans, DBSCAN, Hierarchical — compared & tuned

02 Interactive Map

Folium map — all 62 counties colour-coded by zone

03 Algorithm Comparison

Silhouette & Davies-Bouldin metrics for all three

04 Business Zones

3 actionable zones with strategic targeting advice

05 Charts & EDA

10 deep-dive EDA visualisations with full insights

06 Project Report

Full methodology, code, results documented

DATA CLEANING — NULL HANDLING CHART

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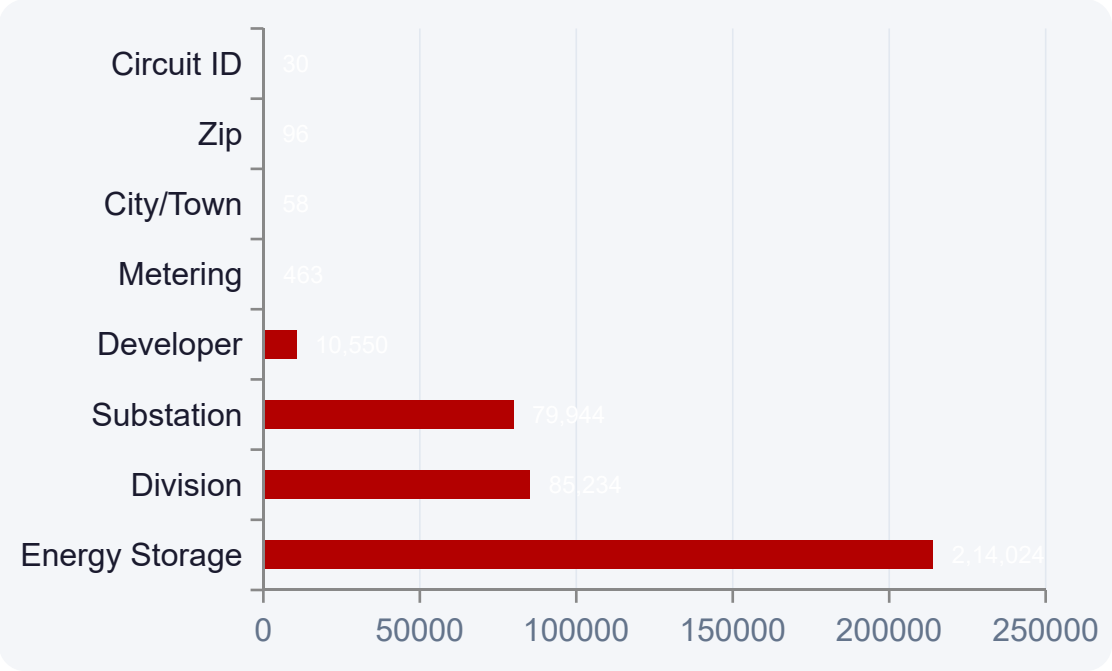
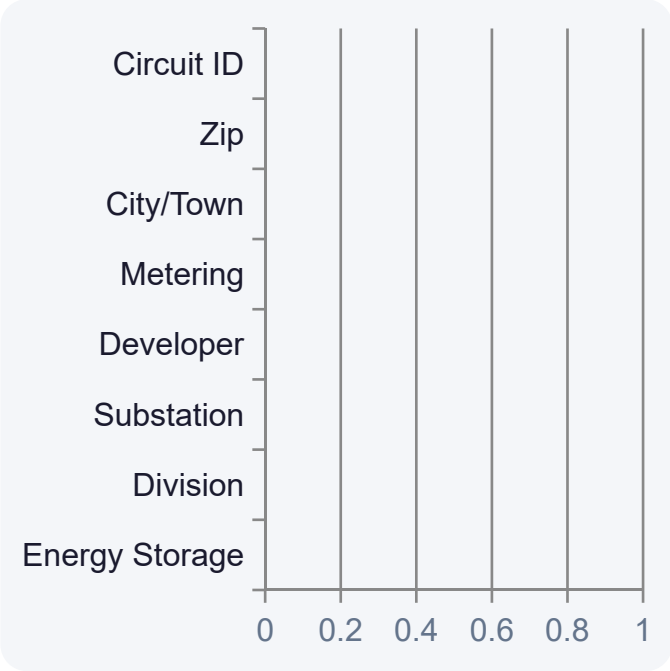
Chart 1: Null Count Before vs After Preprocessing





Energy Storage: 214,024 nulls (98.1%) → filled with 0 — 98% missing = no battery installed. Zero is the correct business value.

Division & Substation: 85K / 80K nulls → filled with 'Unknown' — Missing flags created BEFORE filling to preserve original missingness info.

All nulls resolved → 0 remaining nulls — Result: Clean dataset with 218,115 records ready for analysis.

EDA — ENERGY PRODUCTION DISTRIBUTION CHART

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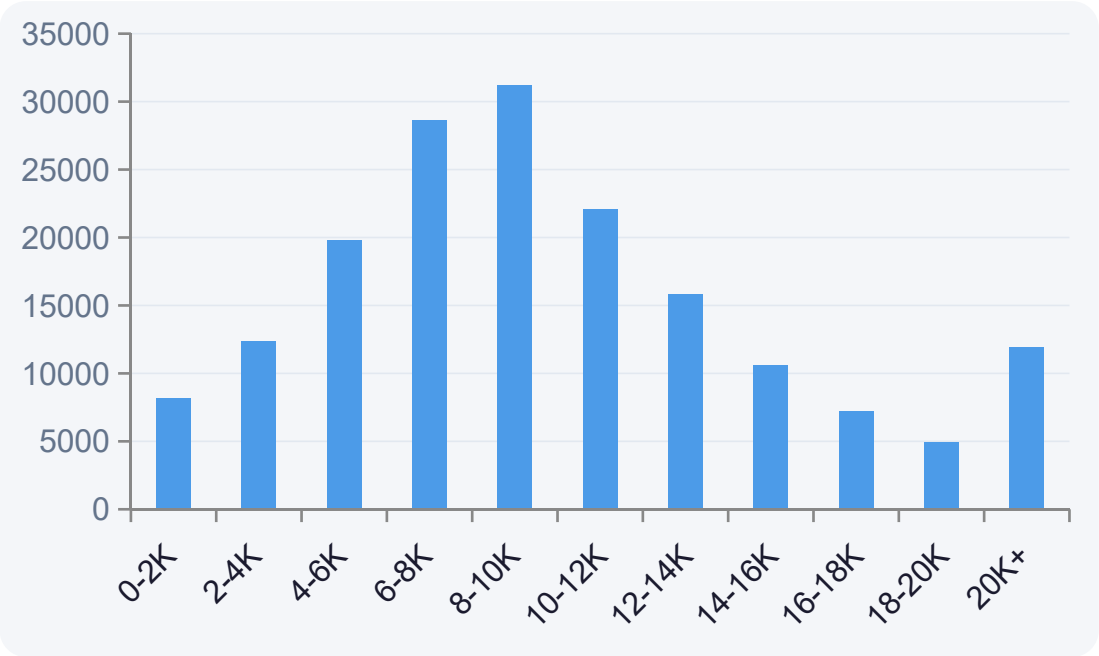
Chart 2: Distribution of Annual PV Energy Production (95th Percentile View)



14 kWh
Minimum

51M kWh
Maximum

28,768
Mean

8,364
Median

42.36
Skewness

Market is Mostly Small Systems: Most projects produce 8K–10K kWh. Mean (28,768) >> Median (8,364) because 3 utility-scale farms in Suffolk (32M–51M kWh each) pull the mean up dramatically.

Extreme Skewness = 42.36: After 15,000 kWh bars become very short. Very few projects produce above 15K kWh. Skew confirms heavy right tail from utility-scale outliers.

Business Decision: Target residential areas for consistent volume. Separately pursue utility-scale opportunities in Suffolk-type regions for maximum energy impact.

TOP COUNTIES BY ENERGY + YEARLY GROWTH TREND

Chart 3: Top 15 Counties — Total Energy Production

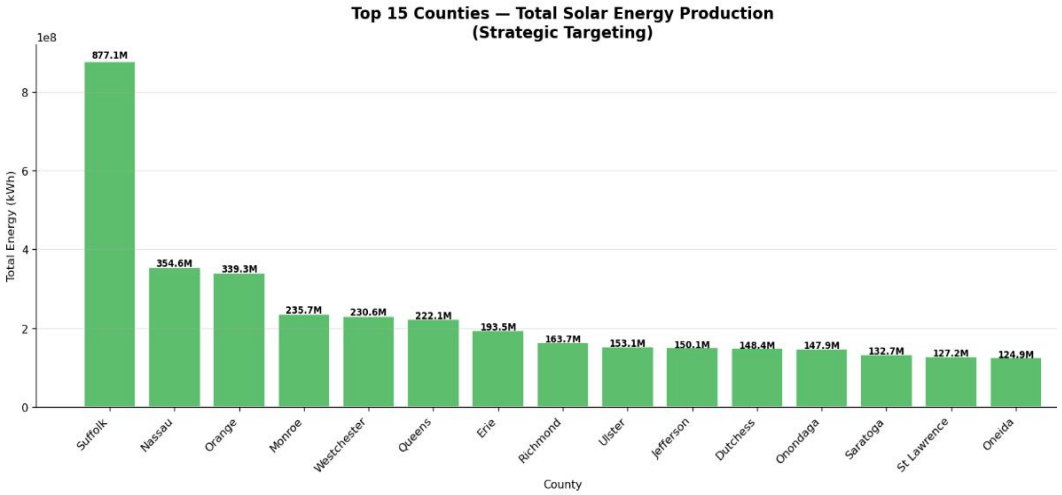
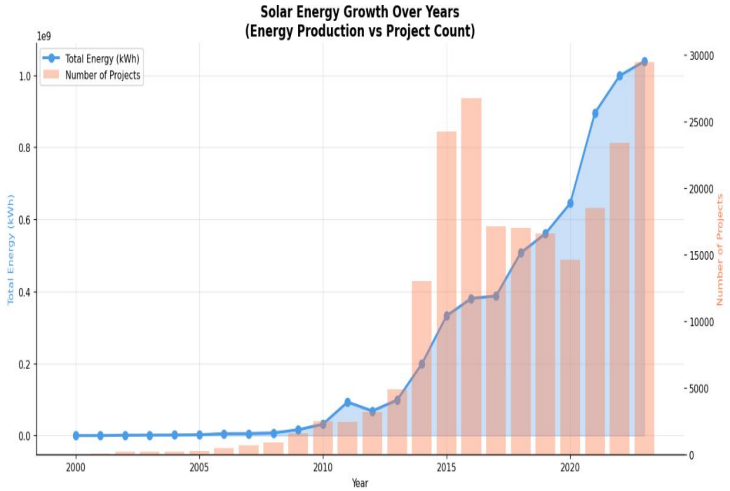


Chart 4: Yearly Growth Trend — Projects vs Energy



Suffolk Dominance: 877M kWh = 2.5× Nassau (#2). All 3 massive utility-scale farms (27K–43K kWdc each) are in Suffolk. Suffolk has BOTH residential + utility-scale market.

2023 = ALL-TIME HIGH: 29,452 new projects. After 2017–2020 cooldown (policy changes), market fully recovered. Blue line = energy NEVER drops — existing projects keep producing.

Tier 1 — Immediate Priority

Suffolk, Nassau, Orange → Deploy sales team first. Highest ROI. Market already mature.

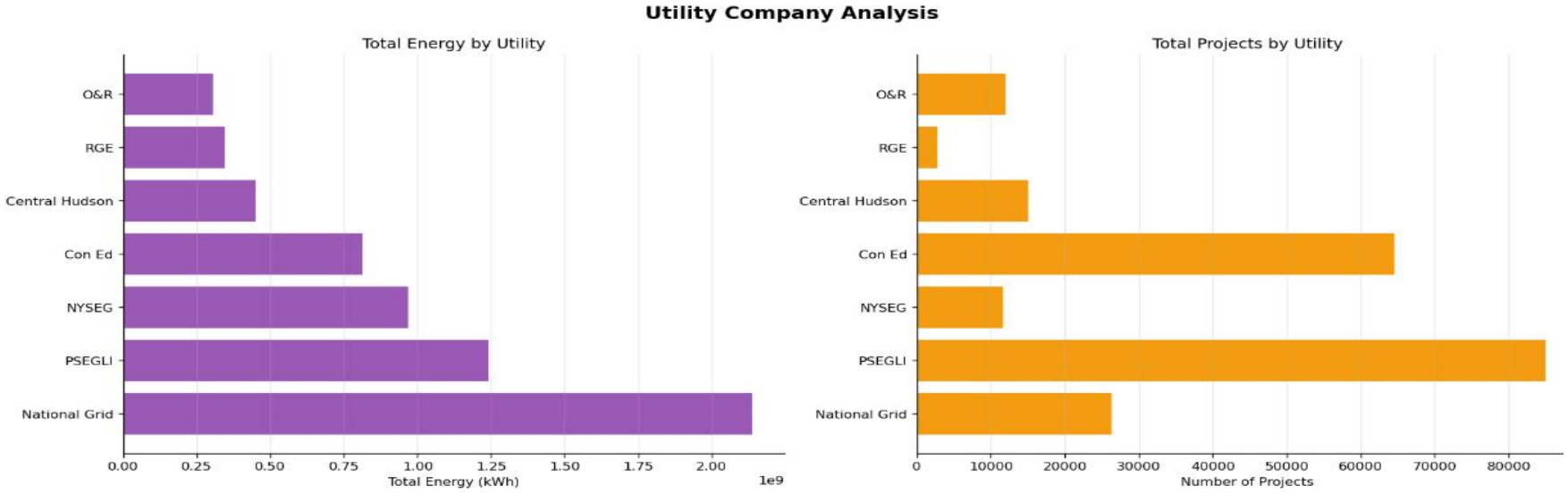
Tier 2 — Secondary Expansion

Monroe, Westchester, Queens, Erie → Strong market. Good pipeline.

Tier 3 — Growth Opportunity

Jefferson, Onondaga, Albany, Saratoga → Growing market. Less competition.

Chart 5: Utility — Energy & Projects



Exact Numbers for Chart 4:

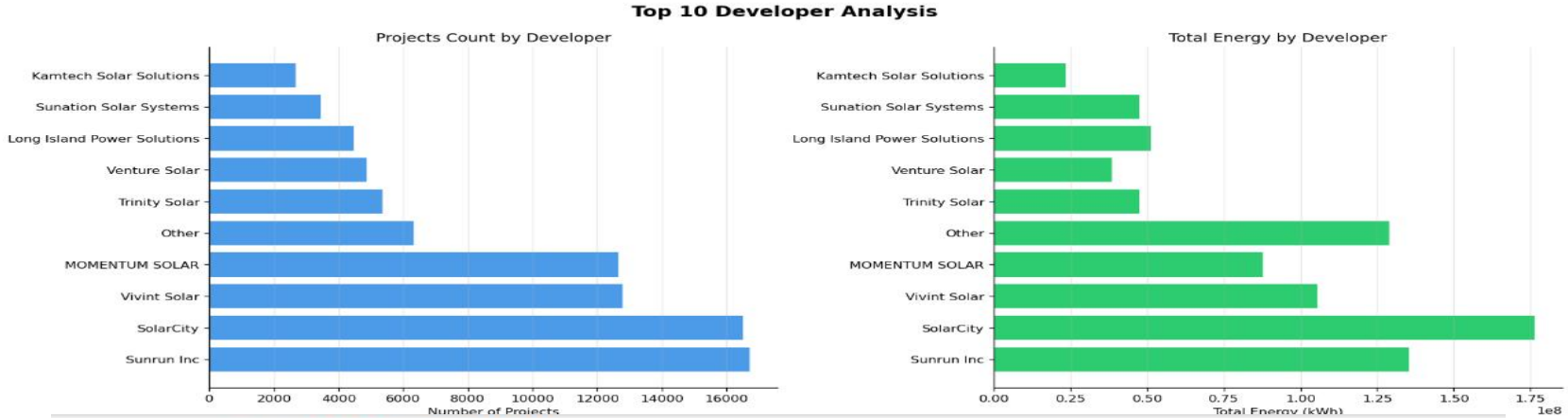
Utility	Total_Energy	Total_Projects
National Grid	2140480498	26400
PSEGLI	1243127779	85110
NYSEG	970550131	11773
Con Ed	816434954	64709
Central Hudson	450707127	15106
RGE	345663301	2865
O&R	307829295	12152

National Grid: Highest energy but few projects. Avg 81,079 kWh/project = 10× typical. Serves large commercial areas.

PSEGLI: Most projects (85,110) but small avg 14,607 kWh. Dominated by residential rooftops on Long Island.

DEVELOPER ANALYSIS CHARTS

Chart 6: Developer- Project count & Total Energy



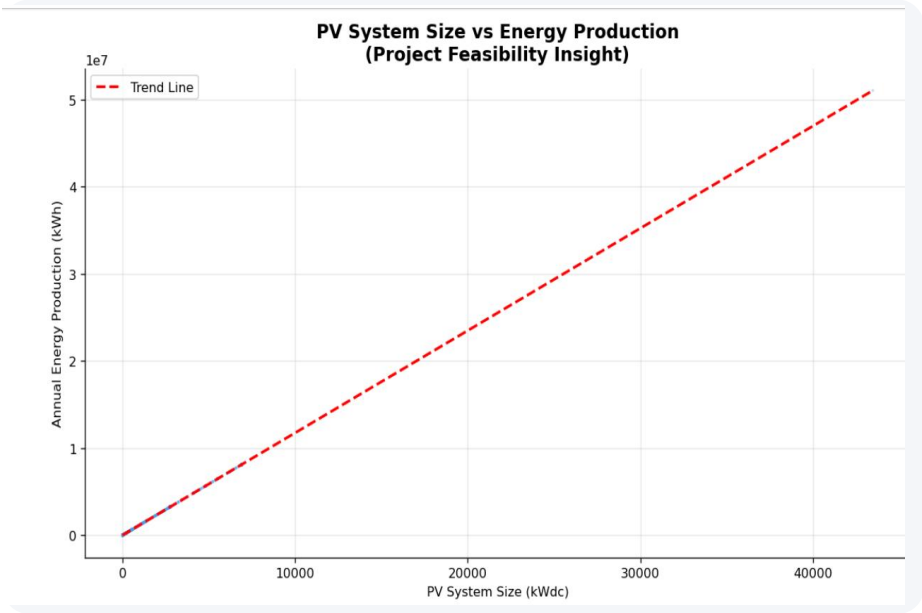
Top 10 Developers:

Developer	Total_Energy	Total_Projects
Sunrun Inc	135557205	16748
SolarCity	176622068	16527
Vivint Solar	105824739	12822
MOMENTUM SOLAR	88002791	12689
Other	129253016	6347
Trinity Solar	47591683	5391
Venture Solar	38795217	4884
Long Island Power Solutions	51456141	4476
Sunation Solar Systems	47668573	3463
Kamtech Solar Solutions	23657627	2703

SolarCity 32% more efficient than Sunrun: 10,686 vs 8,096 kWh avg/project despite fewer projects.

PV SYSTEM SIZE vs ENERGY + STORAGE ADOPTION CHARTS

Chart 7: PV System Size vs Energy Production



8,669 kWh
Small (≤15 kWdc)

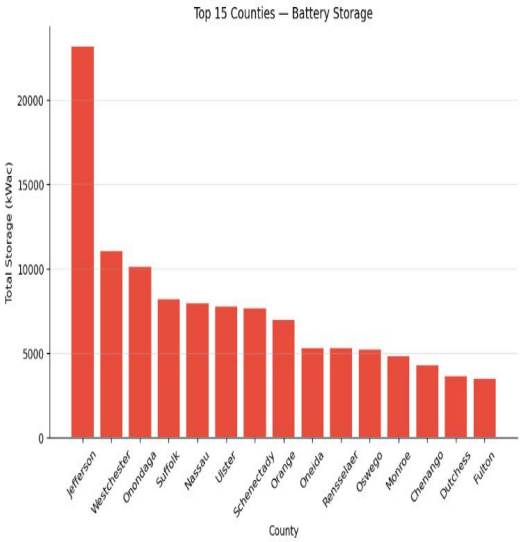
56,845 kWh
Medium (15–1000)

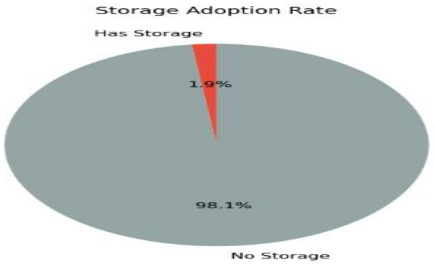
4.68M kWh
Large (>1000 kWdc)

1.0000
Correlation

Large = 540× more energy than small. Correlation 1.0 = zero risk.

Chart 8: Energy Storage Adoption





Only 1.9% adoption
98.1% of projects have NO storage. 214,024 existing solar customers = potential battery upsell market.

Jefferson leads at 23,200 kWac
Nearly double #2 Westchester (11,094 kWac). Jefferson was NOT in top energy counties — storage is an INDEPENDENT market.

Suffolk & Nassau storage gap
#1 and #2 energy producers have low storage adoption(8247 kWac & 7994 kWac)— massive existing customer base for storage upselling.

CORRELATION HEATMAPS — FEATURE SELECTION CHARTS

Chart 9a: Numeric Feature Correlation (for clustering feature selection)

1. Perfect Multicollinearity Detected
PV DC Size, PV AC Size, and Annual Energy Production are perfectly correlated (r = 1.00) — they carry identical information. Larger systems simply produce more energy.

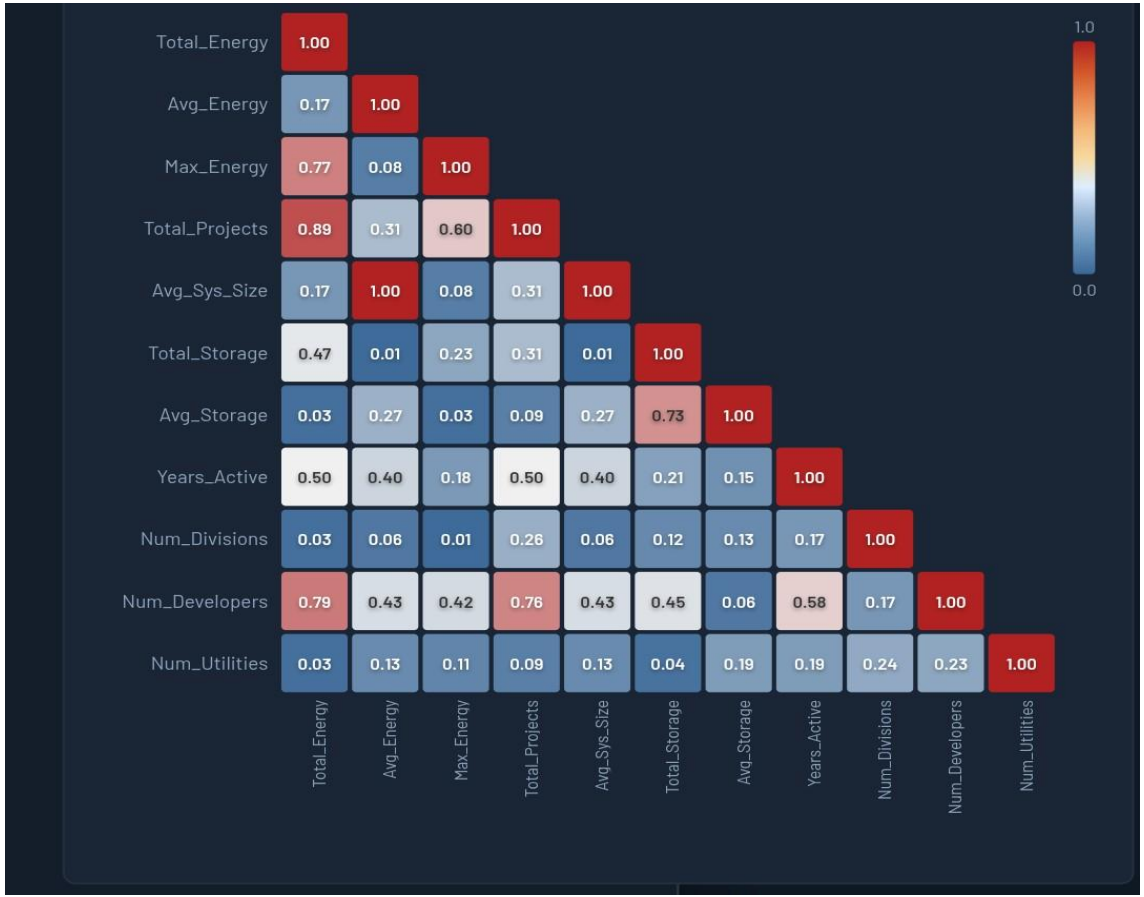
2. Storage Adoption is Independent
Battery storage shows only weak correlation (r = 0.20) with system size and energy output, confirming it operates as a distinct market segment.

3. Year of Installation is Irrelevant
Near-zero correlation (r = 0.03) indicates installation year has no meaningful impact on system size or energy production.

Chart 9b: Categorical Correlation — Cramer's V

- Utility and Division carry almost identical information at 0.99 correlation making one completely redundant. County is the most meaningful geographic feature and should be the primary clustering unit. Metering Method is the only categorical feature that adds truly independent information.

Chart 10: County-Level Feature Correlation (Step 9 — after aggregation)



What Happened:

* Highly correlated features carrying duplicate information were identified and removed.

Results:

* Avg_System_Size dropped
* Correlated 1.00 with Avg_Energy
* 10 features kept for next step

Key Observations From Heatmap:

Total_Energy and Total_Projects show 0.89 correlation which is high but below the 0.90 threshold so both were correctly kept. Avg_Energy and Avg_System_Size show 1.00 correlation confirming they carry identical information. Num_Developers and Total_Projects show 0.76 correlation meaning counties with more projects attract more developers which makes business sense. Num_Divisions and Num_Utilities show low correlation with energy metrics confirming they add independent information

Business Insight:

Removing Avg_System_Size prevents the clustering algorithm from being biased toward system size twice. The remaining 10 features provide a clean non-redundant view of each county's solar profile.

SILHOUETTE SCORE FEATURE TESTING


```
Testing feature combinations...

Top 10 Feature Combinations:
                                                   Columns  Num_Cols  Silhouette
          ['Total_Projects', 'Avg_Storage', 'Num_Developers']         3      0.5312
           ['Total_Projects', 'Total_Storage', 'Avg_Storage']         3      0.5212
               ['Max_Energy', 'Total_Storage', 'Avg_Storage']         3      0.5162
               ['Avg_Energy', 'Max_Energy', 'Total_Projects']         3      0.5073
            ['Total_Energy', 'Total_Projects', 'Avg_Storage']         3      0.5040
                ['Total_Energy', 'Max_Energy', 'Avg_Storage']         3      0.5021
                  ['Avg_Energy', 'Max_Energy', 'Avg_Storage']         3      0.5005
['Total_Energy', 'Avg_Energy', 'Max_Energy', 'Total_Projects']         4      0.4984
           ['Max_Energy', 'Total_Projects', 'Num_Developers']         3      0.4962
             ['Total_Energy', 'Avg_Energy', 'Total_Projects']         3      0.4899

🏆 Best Math Features  : ['Total_Projects', 'Avg_Storage', 'Num_Developers']
🏆 Best Silhouette     : 0.5312

 Final Business Features : ['Total_Energy', 'Total_Projects', 'Avg_Storage']

WHY THESE 3 FEATURES:
→ Total_Energy   : PRIMARY  goal
                   Cluster by energy production
→ Total_Projects : Market size per county
                   More projects = bigger market
→ Avg_Storage    : Future opportunity indicator
                   Storage adoption per county
                   Independent from energy
                   confirmed by correlation

WHY NOT OTHERS:
→ Avg_Energy    : Kept after Step 9 correlation check, but excluded here at Step 10 because it carries the same directional information as Total_Energy
in the county-aggregated dataset. Removing it avoids double-counting the same signal
→ Max_Energy    : Driven by those 3 Suffolk outliers
→ Avg_System_Size: Correlated with energy features
→ Years_Active  : All counties active 20+ years
                  Low variation between counties
→ Num_Divisions : Administrative boundary
                  Not energy performance metric
→ Num_Utilities : Most counties served by 1-2
                  Low variation
→ Metering_encoded: All counties = NM dominant
                    Zero variation confirmed
```

- **What Happened:**

All combinations of 3 to 6 features were tested to find which combination gives best clustering quality

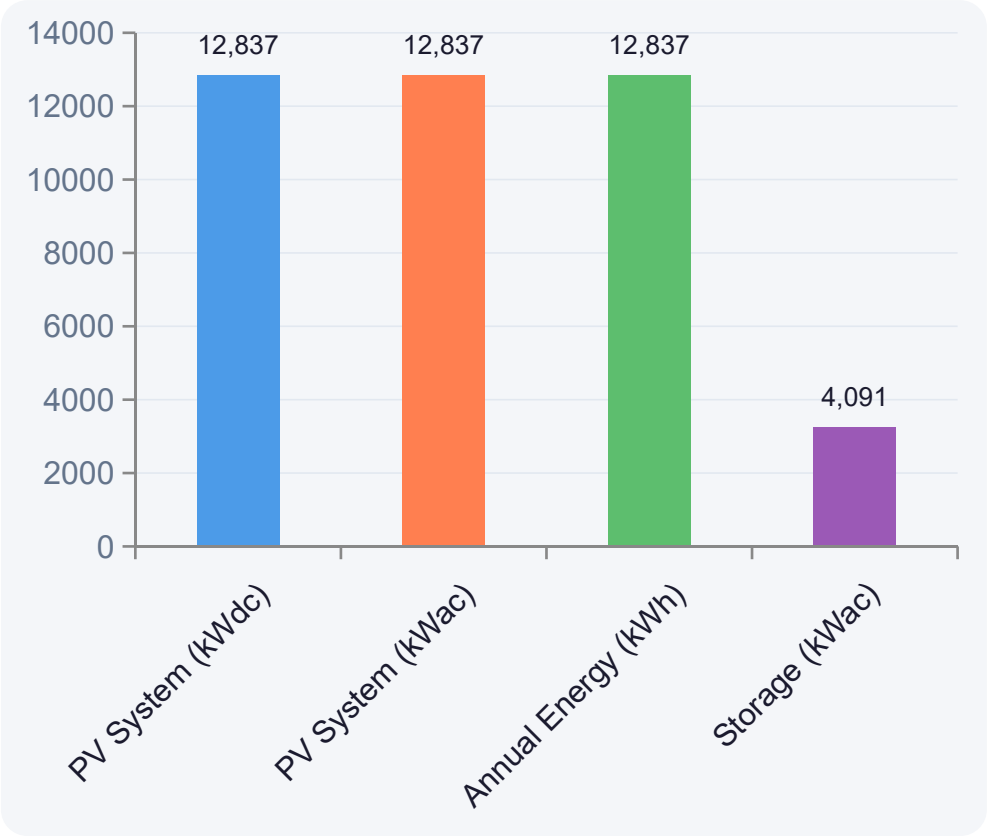
- **Results:**

* Best Math Combination:
* Total_Projects + Avg_Storage + Num_Developers
* Silhouette = 0.5312

- **Business Justified Combination:**

* Total_Energy + Total_Projects + Avg_Storage
* Silhouette = 0.5040

Chart 11: Outlier Counts per Feature



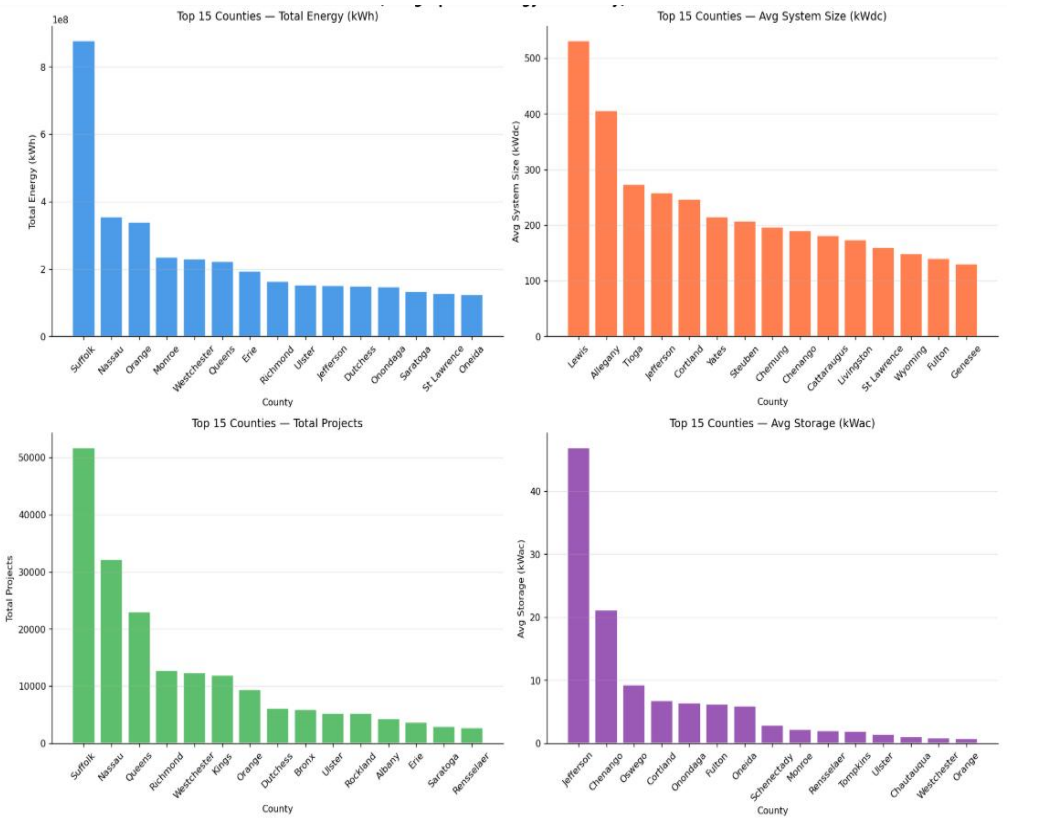
Log Transform Skewness Reduction

Feature	Before Log	After Log	Change
Total_Energy	4.38	-1.35	✓ Reduced
Total_Projects	4.23	0.78	✓ Reduced
Avg_Storage	5.72	2.40	✓ Reduced

12,837 outliers = real utility-scale commercial projects. Must KEEP (not remove) but transform using np.log1p() to prevent them from dominating distance calculations.

COUNTY LEVEL SUMMARY

Chart 12



Top 15 Counties — Total Energy (kWh)
Suffolk dominates total energy production by a wide margin, nearly double Nassau (2nd place). There's a steep drop-off after the top 2, with the remaining counties clustering in a relatively narrow range — suggesting energy production is highly concentrated in a few counties.

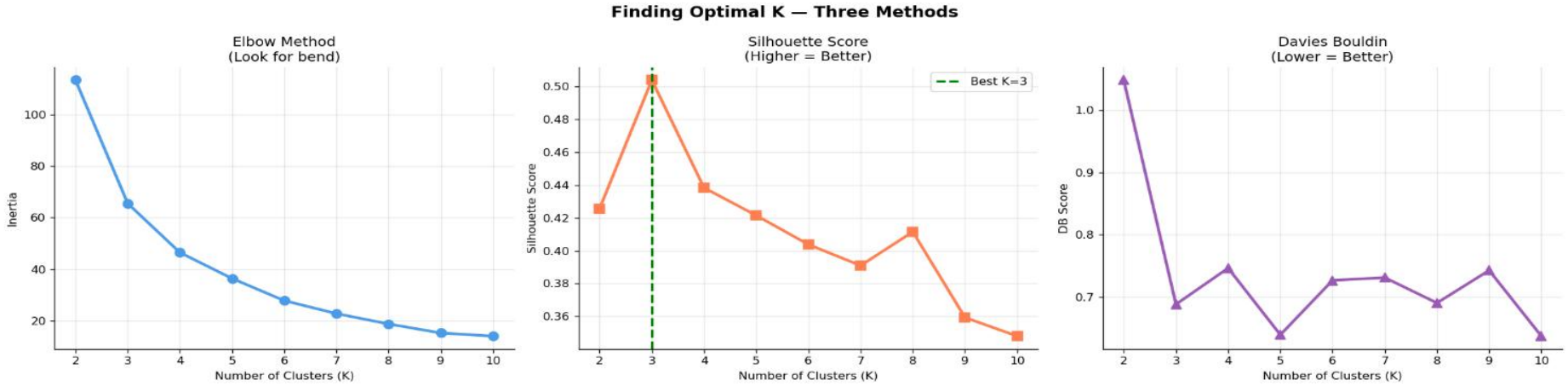
Top 15 Counties — Avg System Size (kWdc)
Lewis leads significantly in average system size, followed by Allegany and Tioga. Interestingly, these are mostly rural/upstate NY counties — not the same high-volume counties from the energy chart — indicating that larger systems are installed in less densely populated areas, likely due to available land.

Top 15 Counties — Total Projects
Suffolk and Nassau again top the list, but Queens, Richmond, and Westchester follow closely. This suggests high project volume in urban/suburban downstate counties, even if individual system sizes are smaller. Volume is driven by population density and rooftop solar adoption.

Top 15 Counties — Avg Storage (kWac)
Jefferson leads by a large margin in battery storage capacity, with Chautauqua and Oswego following. These are mostly rural upstate

KEY: Jefferson is most strategically balanced county — High Avg Storage (46.87 kWac), 4th highest Avg System Size (258 kWdc), top 10 energy county. Lewis has highest avg system size but low project count (not top energy).

FINDING OPTIMAL K — THREE METHOD CHARTS



K	Inertia	Silhouette ↑	Davies-Bouldin ↓	Verdict
K=2	113.29	0.4257	1.0481	
K=3 ★	**65.51**	**0.5040 ★**	**0.6882 ★**	**BEST**
K=4	46.49	0.4384	0.7463	
K=5	36.43	0.4215	0.6393	
K=6	27.85	0.4038	0.7267	

Elbow Method

Inertia drops sharply 113→66 at K=3 then flattens. Clear elbow visible at K=3.

Silhouette Score

Peaks at K=3 (0.5040) then consistently declines through K=10. Unambiguous maximum.

Davies-Bouldin

Lowest score at K=3 (0.6882) = best cluster separation. Confirms both other methods.

CONCLUSION: K=3 unanimously confirmed by all three methods → High / Medium / Low Production Zones

K-MEANS CLUSTERING CHARTS

Chart 14: KMeans — Projects vs Energy (Cluster View)

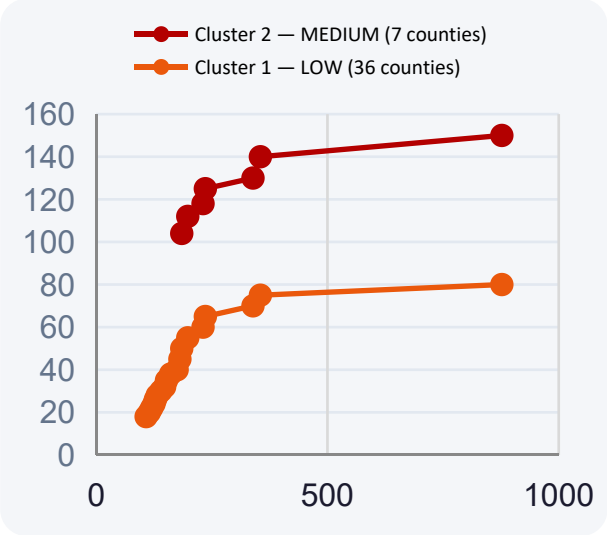
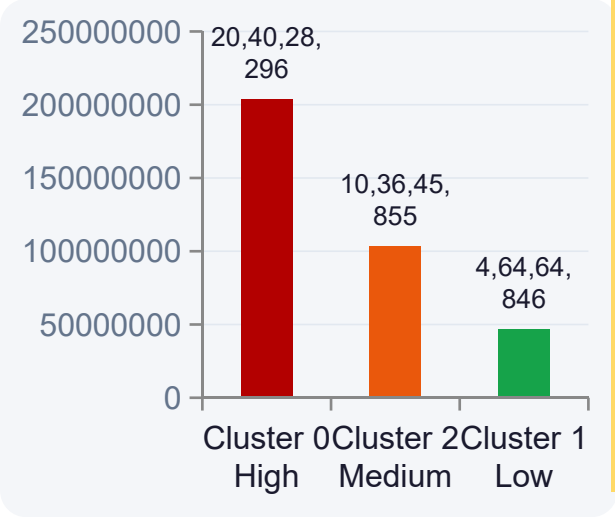


Chart 15: KMeans — Avg Energy per Cluster

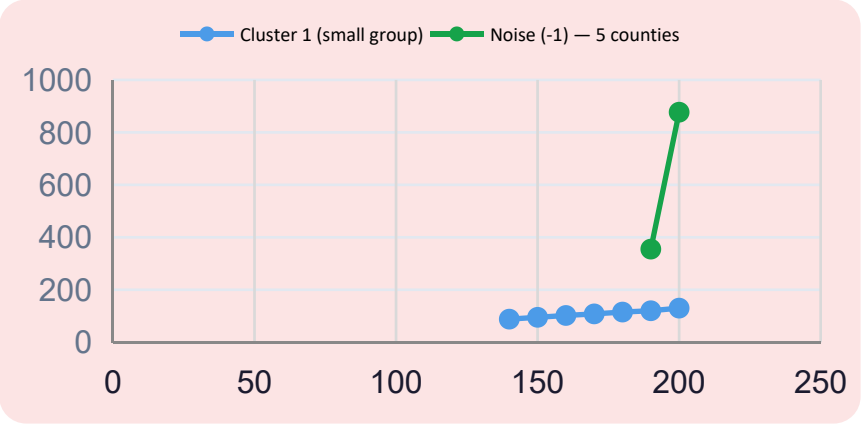


NOTE: KMeans cluster numbers (0,1,2) are arbitrary labels. Final zones use the tuned Hierarchical model — KMeans=19/7/36, Hierarchical=18/7/37. The 1-county difference is Columbia: KMeans placed it in High (Cluster 0) because its 91.9M kWh total energy crossed the centroid threshold. Hierarchical (Ward linkage) reclassified it to Low Zone because Columbia has only 1,476 projects and 0.58 kWac average storage — Ward linkage minimises within-cluster variance, so Columbia's low project count and low storage made it a statistically closer fit for the Low Zone cluster.

Cluster 0 — HIGH (19 counties) **204M kWh avg | 19 Counties**	Albany, Bronx, Columbia, Dutchess, Erie, Kings, Monroe, Nassau, Orange, Queens, Rensselaer, Richmond, Rockland, Saratoga, Schenectady, Suffolk, Tompkins, Ulster, Westchester
Cluster 2 — MEDIUM (7 counties) **104M kWh avg | 7 Counties**	Chenango, Cortland, Fulton, Jefferson, Oneida, Onondaga, Oswego
Cluster 1 — LOW (36 counties) **46M kWh avg | 36 Counties**	Allegany, Broome, Cattaraugus, Cayuga, Chautauqua, Chemung, Clinton, Delaware, Essex, Franklin, Genesee, Greene, Hamilton', 'Herkimer', 'Lewis', 'Livingston', 'Madison', 'Montgomery', 'New York', 'Niagara', 'Ontario', 'Orleans', 'Otsego', 'Putnam', 'Schoharie', 'Schuyler', 'Seneca', 'St Lawrence', 'Steuben', 'Sullivan', 'Tioga', 'Warren', 'Washington', 'Wayne', 'Wyoming', 'Yates.

DBSCAN (WHY IT FAILED) + HIERARCHICAL CLUSTERING CHARTS

Chart 16: DBSCAN — Failed Clustering



0.3335	3.6470	2	5
Silhouette	Davies-Bouldin	Clusters Found	Noise Points
Poor (<0.4)	Very High=Bad	Expected 3	Excluded

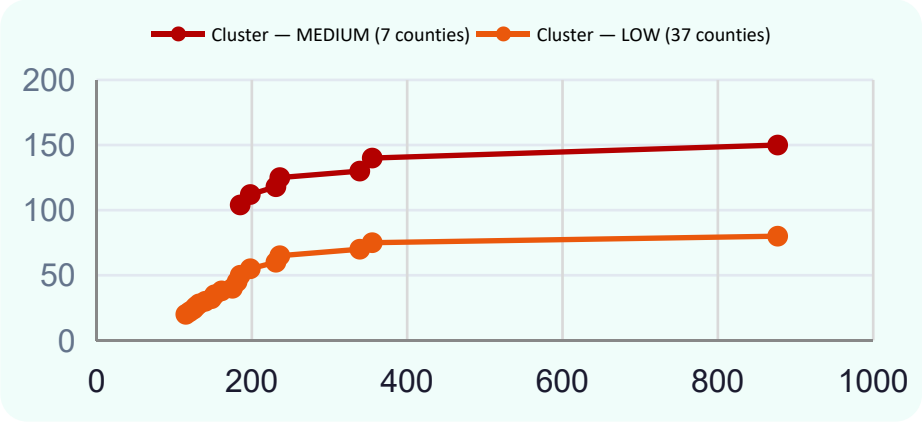
✗ 62 data points too few for dense clusters. DBSCAN needs large point clouds.

✗ Uniform distribution after StandardScaler — no natural density pockets in 3D space.

✗ 5 counties flagged as noise = excluded entirely. Unacceptable for 100% coverage needed. ✗ Tested 36 parameter combinations (eps 0.3→3.0, min_samples 2→5) — no setting achieves silhouette ≥ 0.50 or finds exactly 3 clusters.

VERDICT: DBSCAN ELIMINATED

Chart 17: Hierarchical Clustering Scatter (Ward, K=3)



0.5065	0.6813	18/7/37
Silhouette	Davies-Bouldin	Counties
Better than KMeans	Better than KMeans	High/Med/Low

✓ No random init sensitivity — same result every run.

✓ Dendrogram shows 3 natural groups at Ward distance 12 separation.

✓ 7-county Medium group = most distinct & compact cluster in dataset.

WINNER: Hierarchical (Ward, K=3)

ALGORITHM COMPARISON CHARTS + FINE-TUNING CHART

Chart 18: Algorithm Comparison — Silhouette Score

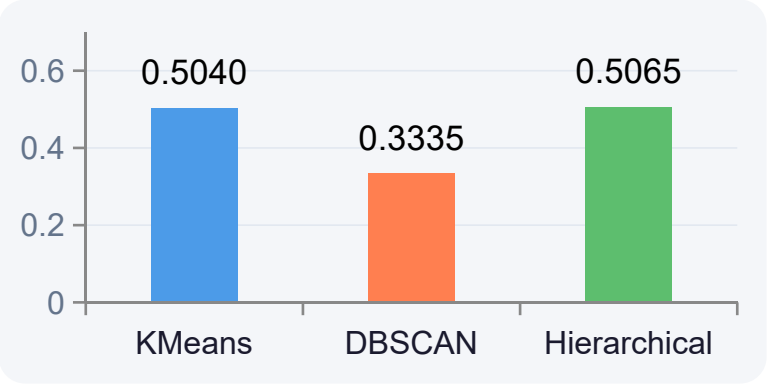


Chart 19: Algorithm Comparison — Davies-Bouldin

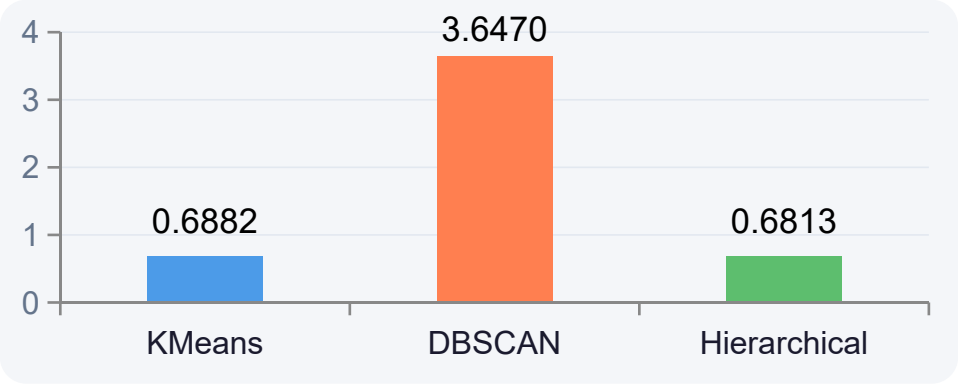
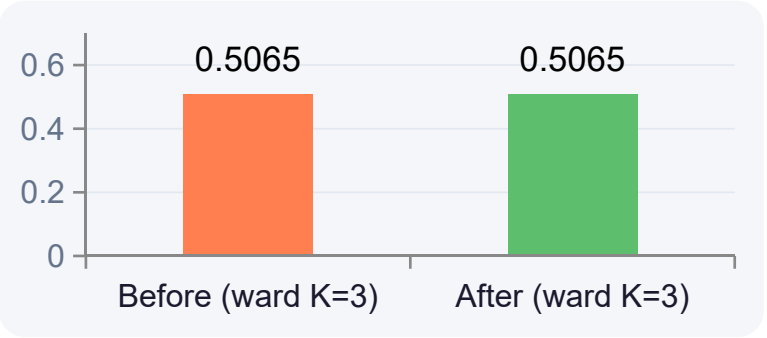


Chart 20: Fine-Tuning — Before vs After



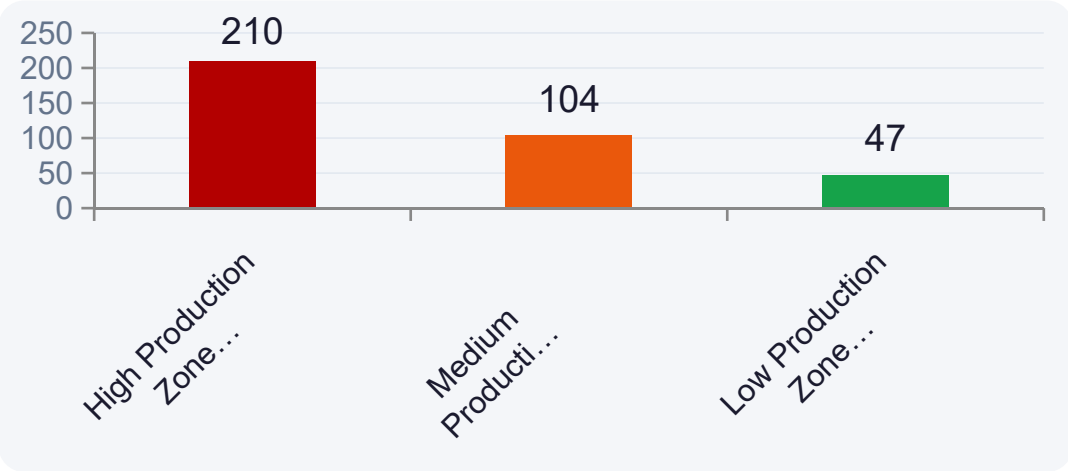
Fine-Tuning Results (12 Combinations)

Configuration	Silhouette	D-B	Verdict
ward + K=3 ★	**0.5065**	**0.6813**	**★ BEST — original**
complete + K=4	0.5053	0.5787	Close, not best
complete + K=3	0.4978	0.7274	Below original
average + K=3	0.4835	0.4466	Below original
ward + K=4	0.4393	0.7463	Below original
single + K=3	0.4722	0.3025	Poor

FINAL MODEL: Hierarchical (Ward, K=3) | Silhouette: 0.5065 | Davies-Bouldin: 0.6813 | Fine-tuning confirmed: ward+K=3 was ALREADY optimal

ZONE LABELLING — ENERGY PER ZONE BOXPLOT CHART

Chart 21: Energy Production per Zone (Zone Boxplot)



Boxplot Reading Guide

High Zone: IQR 100M–350M kWh. Suffolk = extreme outlier at 877M (visible in scatter plots). Wide box confirms high energy diversity within zone.

Medium Zone: Compact tight box 50–150M kWh. 7 counties all perform similarly — consistent and distinct group. Most internally homogeneous zone.

Low Zone: Values concentrated near zero. 37 counties with similar small-scale rural profiles. Majority of NY counties are here.

ZONE 1 (Cluster 1)— HIGH PRODUCTION

18 Counties

210M kWh avg / county

3.78B kWh total

10,897 avg projects

0.79 kWac avg storage

→ NYC metro, Long Island, Hudson Valley. Immediate priority — deploy sales team first. Suffolk & Nassau = primary targets.

ZONE 2(Cluster 2) — MEDIUM PRODUCTION

7 Counties

104M kWh avg / county

725M kWh total

658 avg projects

14.67 kWac avg storage ★

→ Chenango, Cortland, Fulton, Jefferson, Oneida, Onondaga, Oswego. Storage leader — 18× more battery than Zone 1.

ZONE 3 (Cluster 0)— LOW PRODUCTION

37 Counties

47M kWh avg / county

1.76B kWh total

469 avg projects

0.26 kWac avg storage

→ Upstate NY — rural and mid-size counties. Growth opportunity — lowest competition, early mover advantage.

BUSINESS RECOMMENDATIONS & FINAL SUMMARY

TIER 1

Immediate Sales Deployment

Zone 1 — Suffolk, Nassau, Orange, Monroe, Westchester. Highest ROI. Mature market with proven demand and established infrastructure.

TIER 2

Secondary Market Expansion

Zone 1 secondary — Queens, Erie, Richmond, Saratoga. Strong pipeline and good growth potential after Tier 1 captured.

TIER 3

Storage Market Entry

Zone 2 — Jefferson leads battery storage (46.87 kWac avg). Early mover advantage in rapidly growing solar+storage market.

TIER 4

Long-Term Upstate Growth

Zone 3 — 37 counties with lowest competition. Position now for market penetration before saturation arrives.

FINAL SUMMARY

218,115 records	62 counties	3 actionable zones	Best: Hierarchical (Ward, K=3)	Silhouette: 0.5065	Davies-Bouldin: 0.6813

Thank You